

Skills Summary

Graphing Basic Polar Curves from Equations

Use this reference while completing your worksheet or when studying.

Skill 1 — Plotting one point of a polar curve

Rule: a polar equation is a recipe. Feed it an angle θ , it returns a directed distance r . Plot the point by facing along the ray θ and walking r units — forwards if $r > 0$, backwards through the pole if $r < 0$.

Example: Find r on $r = 2 + 3 \sin \theta$ at $\theta = \frac{\pi}{6}$.

Step 1. One angle in, one distance out — substitute and evaluate, no graphing needed yet.

Step 2. $\sin \frac{\pi}{6} = \frac{1}{2}$, so $r = 2 + 3 \left(\frac{1}{2} \right)$.

$\Rightarrow r = \frac{7}{2}$, so the point sits 3.5 units out along the $\pi/6$ ray.

Try it: Find r on $r = 5 + \cos \theta$ at $\theta = \frac{\pi}{3}$.

check: $r = \frac{11}{2}$

Skill 2 — Circles and lines in polar form

The three shapes to recognise on sight:

- $r = k$ — circle of radius k centred at the pole.
- $\theta = k$ — straight line through the pole at angle k .
- $r = a \cos \theta$ — circle of diameter a resting on the polar axis; $r = a \sin \theta$ — the same circle rotated onto the vertical axis.

Example: Locate a point of the circle $r = 4 \sin \theta$ at $\theta = \frac{\pi}{6}$.

Step 1. The shape is already known to be a circle, so one substituted point is enough to pin its position.

Step 2. $\sin \frac{\pi}{6} = \frac{1}{2}$, so $r = 4 \left(\frac{1}{2} \right)$.

$\Rightarrow r = 2$ at $\theta = \pi/6$; the circle's far point is at $\theta = \pi/2$.

Try it: Find r on the circle $r = 8 \cos \theta$ at $\theta = \frac{\pi}{3}$.

check: $r = 4$

Skill 3 — Limacons and cardioids

Form: $r = a \pm b \cos \theta$ or $r = a \pm b \sin \theta$. Compare a with b :

- $a = b$ — cardioid (heart), reaches the pole exactly once.
- $a > b$ — dimpled or convex limaçon, never reaches the pole.
- $a < b$ — limaçon with an inner loop, traced where $r < 0$.

Example: Find the largest r on $r = 3 + 2 \cos \theta$.

Step 1. r is largest where the cosine is largest, and $\cos \theta$ peaks at 1 when $\theta = 0$.

Step 2. $r = 3 + 2(1)$ at $\theta = 0$.

$\Rightarrow r_{\max} = 5$, so the curve stretches farthest along the polar axis.

Try it: Find r on $r = 4 - 3 \sin \theta$ at $\theta = \frac{\pi}{2}$.

check: $r = 1$

Skill 4 — Rose curves

Form: $r = a \cos(n\theta)$ or $r = a \sin(n\theta)$. Each petal has length $|a|$; the petal count is n when n is odd and $2n$ when n is even. The curve returns to the pole wherever the sine or cosine is zero, and those angles separate the petals.

Example: Find the tip length of a petal of $r = 4 \sin(2\theta)$.

Step 1. A petal tip is where $|r|$ is largest, so look for the angle that makes the sine equal 1.

Step 2. $2\theta = \frac{\pi}{2}$ gives $\theta = \frac{\pi}{4}$, and $r = 4 \sin \frac{\pi}{2}$.

$\Rightarrow r = 4$, and since $n = 2$ is even this rose has four petals of that length.

Try it: Find r on $r = 6 \cos(3\theta)$ at $\theta = 0$.

check: $r = 6$

(!) **Watch out:** a negative r is not an error. Plot it backwards through the pole — that is exactly how an inner loop and half the petals of an even rose get drawn.